Nandhan Alahari

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EDUCATION

University of Texas at Dallas (UTD) Richardson, TX
B.S. Computer Science. GPA: 4.0 / 4.0. *December 2027*

EXPERIENCE

Software Engineer, BeyondNeuro (UTD EPICS) Aug 2026 - Present
University of Texas at Dallas (UTD) *Richardson, TX*

- Architecting a cognitive wellness web platform for stroke survivors, dementia patients, and caregivers; serving as software engineer and project liaison on a UTD EPICS team partnered with a nonprofit client through biweekly stakeholder syncs.
- Translated clinical priorities into a 4-sprint engineering roadmap spanning interactive cognitive modules, an activity library, local progress tracking, and a PWA with an AI avatar.
- Shipping Sprint 1 of 4: the platform's first cognitive training modules and user onboarding flow; designed the system for WCAG accessibility and role-based access control to serve patients, family caregivers, and healthcare providers.

Research Assistant, Computational Neuroscience Lab (Dr. Zirong Gu) Jun 2026 - Aug 2026
University of Texas at Dallas (UTD) *Richardson, TX*

- Diagnosed and restored a broken production data pipeline (MATLAB) implementing affinity propagation clustering over multi-camera behavioral tracking data (3D pose estimation, acceleration, motion frequency), resolving classification failures that had stalled analysis across 12+ experimental sessions.
- Engineered a Python video-processing pipeline from scratch to auto-generate per-cluster outputs from clustered sensor data; redesigned the core cluster-to-frame mapping algorithm, cutting processing time 60% on 10+ hour recording sessions.

Machine Learning Researcher Jun 2024 - Sep 2024
University of Texas at Dallas (UTD) *Richardson, TX*

- Built MemeQA, a multimodal system (computer vision + NLP) integrating vision-language models to classify sentiment, intent, and context in memes.
- Engineered a data curation pipeline and LoRA fine-tuning workflow with adversarial filtering across a 9K+ question dataset; co-authored the resulting paper under Dr. Vincent Ng.
- Published at ACL 2025, the flagship conference of the Association for Computational Linguistics (one of the top venues in NLP research): **aclanthology.org/2025.acl-long.927**

PROJECTS

NavRisk - 1st Place
EPPS-American Airlines Data Challenge

- Architected and shipped a full-stack web application backed by a RESTful API, surfacing crew-risk scores and scheduling recommendations across delay propagation, duty violations, missed connections, and weather risk; placed 1st out of 50+ teams against graduate and PhD-level competitors.
- Built an XGBoost scoring model and a 36K+ feature engineering pipeline over 6M+ BTS flight records spanning 141 airports and 21 years; integrated a live risk network graph with real-time METAR weather data.

Nestor - Top 5
Goldman Sachs & JSOM Hackathon

- Engineered a portfolio rebalancing engine combining XGBoost return prediction with Mean-Variance Optimization (MVO) to minimize risk, trained on a Kaggle dataset spanning 8,049 tickers (5,884 stocks, 2,165 ETFs).
- Combined real-time Yahoo Finance data with 10 FRED macroeconomic indicators (Treasury yields, CPI, unemployment, VIX), feeding 6 into the XGBoost model; used the Gemini API to translate optimization output into plain-language insights.

Kolana - Best Use of Solana
HackHCC

- Built a voice-driven Solana payment app on a FastAPI backend: Gemini 2.5 Flash parses natural language into payment intent and executes on-chain wallet-to-wallet SOL transfers in roughly 400ms, with ElevenLabs generating real-time voice confirmation.
- Deployed an InsightFace facial recognition model on a DigitalOcean server, storing 512-dimensional face embeddings in MongoDB Atlas with Vector Search for real-time identity resolution against enrolled contacts; learned Solana/blockchain fundamentals from scratch to ship the integration within the hackathon window.

SKILLS

Languages: Python, Java, C++, C, C#, Swift, MATLAB, Assembly, SQL

Frameworks & Tools: React, React Native, Node.js, GitHub Actions, Supabase, Git, Linux, Microsoft Office (Word, Excel, PowerPoint, Teams), Google Workspace (Docs, Sheets, Meet)

Concepts: Machine Learning, Computer Vision, NLP, LLMs, Vector Databases, RESTful APIs, CI/CD, Agile Development, Data Structures & Algorithms